

Social Science Inquiry II

Week 8: P-values and hypothesis testing

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Winter 2026

Loading packages for this class

```
library(ggplot2)
library(estimatr)
library(modelsummary)
library(gridExtra)
set.seed(60637)
config_modelsummary(startup_message = FALSE)

# Make modelsummary output LaTeX via kableExtra (avoids tabular)
options(
  modelsummary_factory_default = "kableExtra",
  modelsummary_factory_latex = "kableExtra",
  modelsummary_factory_html = "kableExtra"
)
```

- Housekeeping.
- Today: interpreting p-values; how hypothesis tests relate to confidence intervals; why *statistical significance* is a shaky decision rule.

Reading in the data

```
file <- "../data/card-krueger.csv"
if (!file.exists(file)) {
  file <- "https://raw.githubusercontent.com/UChicago-pol-methods/SOSC13200-W26/main/data/card-krueger.csv"
}
dat <- read.csv(file, as.is = TRUE)
```

```
head(dat)
```

##	id	nj	d	d_nj	bk	kfc	roys	wendys	co_owned	centralj	southj	pa1	pa2	fte	ft
## 1	1	0	0	0	1	0	0	0	0	0	0	1	0	40.5	30.0
## 2	2	0	0	0	0	1	0	0	0	0	0	1	0	13.8	6.5
## 3	3	0	0	0	0	1	0	0	1	0	0	1	0	8.5	3.0
## 4	4	0	0	0	0	0	0	1	1	0	0	1	0	34.0	20.0
## 5	5	0	0	0	0	0	0	1	1	0	0	1	0	24.0	6.0
## 6	6	0	0	0	0	0	0	1	1	0	0	1	0	20.5	0.0
##	pt	mgrs	wage	meal	hrsopen	bonus	ncalls	status	type	inctime	firstinc	nregs			
## 1	15.0	3	NA	2.6	16	1	0	1	1	19	NA	3			
## 2	6.5	4	NA	4.3	13	0	0	1	1	26	NA	4			
## 3	7.0	2	NA	4.0	10	0	0	1	1	13	0.37	3			
## 4	20.0	4	5.0	3.5	12	1	0	1	1	26	0.10	2			
## 5	26.0	5	5.5	3.3	12	1	0	1	1	52	0.15	2			
## 6	31.0	5	5.0	2.6	12	0	2	1	1	26	0.07	2			

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

Independent variable	Model				
	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap ^a	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership ^b	no	yes	no	yes	yes
4. Controls for region ^c	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls ^d	—	0.34	—	0.44	0.40

Notes: Standard errors are given in parentheses. The sample consists of 357 stores with available data on employment and starting wages in waves 1 and 2. The dependent variable in all models is change in FTE employment. The mean and standard deviation of the dependent variable are -0.237 and 8.825 , respectively. All models include an unrestricted constant (not reported).

^aProportional increase in starting wage necessary to raise starting wage to new minimum rate. For stores in Pennsylvania the wage gap is 0.

^bThree dummy variables for chain type and whether or not the store is company-owned are included.

^cDummy variables for two regions of New Jersey and two regions of eastern Pennsylvania are included.

^dProbability value of joint F test for exclusion of all control variables.

Table 4

	(1)	(2)	(3)	(4)	(5)
New Jersey Dummy	2.326 (1.458)	2.304 (1.454)			
Initial Wage Gap			15.653* (6.178)	14.916* (6.526)	11.979+ (7.035)
Num.Obs.	357	357	357	357	357
AIC	2569.1	2572.6	2566.4	2570.5	2573.9
BIC	2580.8	2599.7	2578.0	2597.7	2616.5
RMSE	8.77	8.71	8.73	8.69	8.63
Controls for chain and ownership	no	yes	no	yes	yes
Controls for region	no	no	no	no	yes

+ $p \leq 0.1$, * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$

Recall our multivariate regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \epsilon$$

- How do we interpret β_0 ? β_1 ? β_K ?

- We observe some data, which we (maybe) assume is randomly sampled from a larger population.
- The model describes the true relationships among the variables.
- But the true population parameters are generally unknown.

- We estimate the parameter values for a given sample, as the values that minimize the sum of squared residuals.

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \cdots + \hat{\beta}_K X_{Ki}$$

- Recall that the residual is defined as:

$$\hat{\epsilon}_i = \hat{Y}_i - Y_i$$

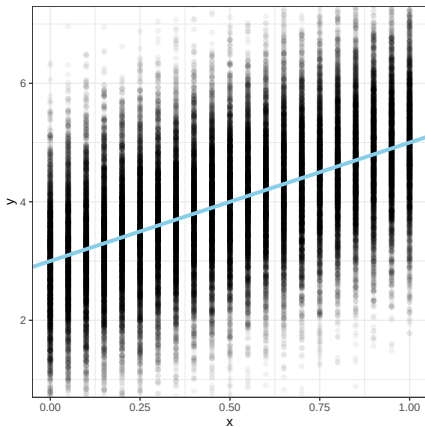
- We will think about our random sample being not just for one variable, but from the joint distribution of $(Y, X_1, X_2, \dots, X_K)$.
- Then each $\hat{\beta}_k$ is also random, with its own sampling distribution.

- We can get a point estimate for each of the parameters, $\hat{\beta}_k$: the coefficients in our linear model.
- We also want to get an estimate of the standard errors of the estimates, $\sqrt{\hat{\text{Var}}[\hat{\beta}_k]}$, to describe how much we think these coefficients vary across samples.

Suppose our true relationship is:

$$Y = 3 + 2X_1 + \epsilon$$

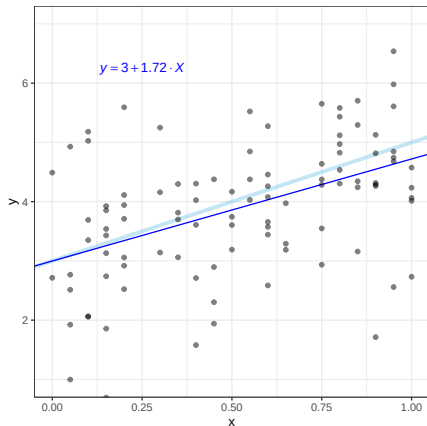
If we were to see the full data:



Suppose our true relationship is:

$$Y = 3 + 2X_1 + \epsilon$$

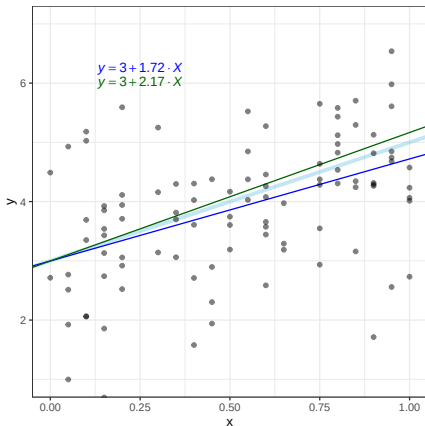
If we only saw 100 observations from the data:



Suppose our true relationship is:

$$Y = 3 + 2X_1 + \epsilon$$

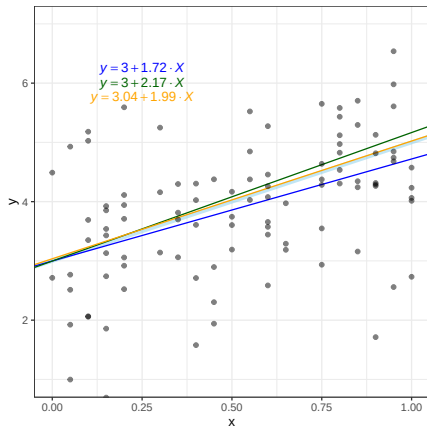
If we only saw 100 observations from the data:



Suppose our true relationship is:

$$Y = 3 + 2X_1 + \epsilon$$

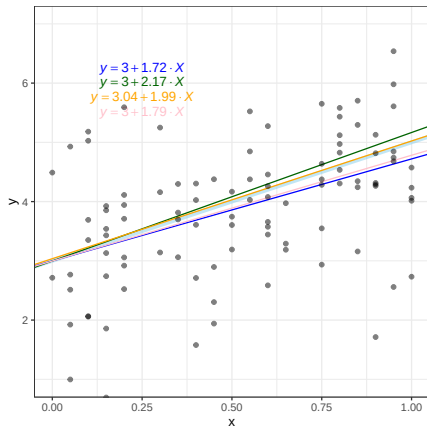
If we only saw 100 observations from the data:



Suppose our true relationship is:

$$Y = 3 + 2X_1 + \epsilon$$

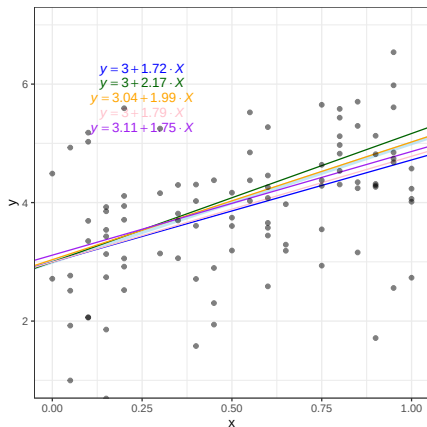
If we only saw 100 observations from the data:



Suppose our true relationship is:

$$Y = 3 + 2X_1 + \epsilon$$

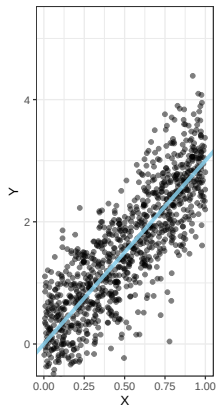
If we only saw 100 observations from the data:



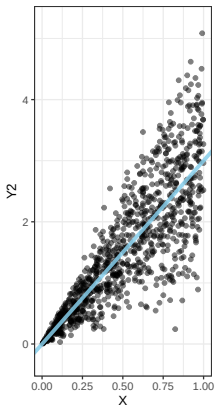
- Each time we sample from the population, we get a slightly different fit for our regression line.
- Our goal is to describe the *variability* in our parameter estimates.
- Here, we have a regression line with just an intercept and a slope. But we could consider the same resampling and fitting procedure for any joint distribution of $(Y, X_1, X_2, \dots, X_K)$,

- We will use **robust** standard errors.
- These standard errors don't require much beyond that our data is i.i.d.: random samples from the same joint distribution.
- "Classical" regression modeling puts much stronger assumptions on the data, including that errors are "homoskedastic;" they don't vary with X

Homoskedastic data



Heteroskedastic data



Using classical standard errors

```
summary(lm(y ~ x, data = data.frame(x = X, y = Y)))$coef[,1:2]
```

```
##              Estimate Std. Error
## (Intercept)    0.022      0.036
## x              2.972      0.062
```

```
summary(lm(y ~ x, data = data.frame(x = X, y = Y2)))$coef[,1:2]
```

```
##              Estimate Std. Error
## (Intercept)    0.006      0.035
## x              3.015      0.060
```

Using robust standard errors

```
broom::tidy(lm_robust(y ~ x, data = data.frame(x = X, y = Y)))[,1:3]
```

```
##           term estimate std.error
## 1 (Intercept)    0.022    0.036
## 2           x     2.972    0.062
```

```
broom::tidy(lm_robust(y ~ x, data = data.frame(x = X, y = Y2)))[,1:3]
```

```
##           term estimate std.error
## 1 (Intercept)    0.006    0.021
## 2           x     3.015    0.063
```

Applied example

Recall:

Pager, D. (2003). The mark of a criminal record.
American Journal of Sociology, 108(5), 937-975.

```
dfp <- data.frame(  
  black = rep(c(0, 1), times = c(300, 400)),  
  record = c(rep(c(0, 1), each = 150),  
             rep(c(0, 1), each = 200)),  
  call_back = c(  
    # whites without criminal records  
    rep(c(0, 1), times = c(99, 51)), # 150  
    # whites with criminal records  
    rep(c(0, 1), times = c(125, 25)), # 150;  
    # - callbacks could be 25 or 26  
    # blacks without criminal records  
    rep(c(0, 1), times = c(172, 28)), # 200  
    # blacks with criminal records  
    rep(c(0, 1), times = c(190, 10)) # 200  
  )  
)
```

- Let's try this with the dfp data, where the outcome Y is `call_back`, regressed on `black` and `record`, interacted.

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{Black}_i + \hat{\beta}_2 \text{Record}_i + \hat{\beta}_3 \text{Black}_i \times \text{Record}_i$$

```
model2 <- lm_robust(call_back ~ black*record, data = dfp)
```


- How do we go about interpreting these coefficients? confidence intervals? p-values?

```
summary(model2)

##
## Call:
## lm_robust(formula = call_back ~ black * record, data = dfp)
##
## Standard error type:  HC2
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|) CI Lower CI Upper  DF
## (Intercept)   0.3400    0.0388   8.76 1.46e-17  0.2638  0.4162 696
## black        -0.2000    0.0459  -4.35 1.55e-05 -0.2902 -0.1098 696
## record       -0.1733    0.0494  -3.51 4.76e-04 -0.2703 -0.0764 696
## black:record   0.0833    0.0573   1.45 1.46e-01 -0.0291  0.1958 696
##
## Multiple R-squared:  0.0771 , Adjusted R-squared:  0.0732
## F-statistic: 18.4 on 3 and 696 DF,  p-value: 1.76e-11
```

Confidence intervals

- A valid confidence interval CI_n for a target parameter θ with coverage $1 - \alpha$

$$P[\theta \in CI_n] \geq 1 - \alpha$$

- If $\alpha = 0.05$, the probability that we construct a random confidence interval such that the estimand θ is in our confidence interval is greater than or equal to 0.95.
- CI_n is a random interval. It is a function of the data we observe.
- θ is a fixed parameter. It does not move.
(In the frequentist view of statistics.)
- If you use valid confidence repeatedly in your work, 95% of the time, your confidence intervals will include the true value of the relevant θ .

- The formula for the 95% confidence interval is:

$$CI_n = \left(\hat{\theta}_n - 1.96 \times \hat{se}, \hat{\theta}_n + 1.96 \times \hat{se} \right)$$

- The 1.96 value tells us how many standard errors away from the mean we need to include in our interval to get valid coverage.
- This formula is based on a *normal approximation*, i.e., we assume the data is going to look like a normal distribution.

- How do we go about interpreting these coefficients? confidence intervals? p-values?

```
confint(model2)
```

```
##           2.5 % 97.5 %  
## (Intercept)  0.264  0.416  
## black       -0.290 -0.110  
## record      -0.270 -0.076  
## black:record -0.029  0.196
```

P-values

Suppose $\hat{\theta}$ is the general form for an estimate produced by our estimator, and $\hat{\theta}^*$ is the value we have actually observed.

P-values

- A two-tailed p-value under the null hypothesis is

$$p = P_0[|\hat{\theta}| \geq |\hat{\theta}^*|]$$

i.e., the probability *under the null distribution* that we would see an estimate of $\hat{\theta}$ as or more extreme as what we saw from the data.

- Confidence intervals and p-values help us get back to testing hypotheses.

Two-sided hypotheses

$$H_0 : \theta = 0$$

$$H_A : \theta \neq 0$$

Note that we are *not* imposing the sharp null of no individual effect here, we're looking at averages.

- Confidence intervals and hypothesis tests have a specific relationship.
- Consider all of the hypotheses that take the form:

$$H_0 : \theta = \theta_0$$

$$H_A : \theta \neq \theta_0$$

- If the calculated two-tailed p -value is less than 0.05, reject the hypothesis.
- If the calculated two-tailed p -value is greater than 0.05, fail to reject the hypothesis.
- The θ_0 for which we would fail to reject the hypothesis lie within the 95% confidence interval.

One way that this is very useful:

- If 0 is outside the 95% confidence interval, we would reject the hypothesis that $\theta = 0$ at $p \leq 0.05$.
- If 0 is outside the 99% confidence interval, we would reject the hypothesis that $\theta = 0$ at $p \leq 0.01$.
- If 0 is outside the 99.9% confidence interval, we would reject the hypothesis that $\theta = 0$ at $p \leq 0.001$.

- How do we go about interpreting these coefficients? confidence intervals? p-values?

```
summary(model2)

##
## Call:
## lm_robust(formula = call_back ~ black * record, data = dfp)
##
## Standard error type: HC2
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|) CI Lower CI Upper DF
## (Intercept)    0.3400    0.0388   8.76 1.46e-17  0.2638  0.4162 696
## black          -0.2000    0.0459  -4.35 1.55e-05 -0.2902 -0.1098 696
## record         -0.1733    0.0494  -3.51 4.76e-04 -0.2703 -0.0764 696
## black:record    0.0833    0.0573   1.45 1.46e-01 -0.0291  0.1958 696
##
## Multiple R-squared:  0.0771 , Adjusted R-squared:  0.0732
## F-statistic: 18.4 on 3 and 696 DF,  p-value: 1.76e-11

round(model2$p.value,5)

## (Intercept)          black          record black:record
##      0.00000      0.00002      0.00048      0.14622
```

Bootstrap estimation

- Another approach to estimating the standard error of an estimate is to use bootstrapping.
- If we fully knew the joint distribution of our population, we would know exactly how to determine the sampling variation of our estimate.
- While we do not, we can *suppose* that the empirical joint distribution produced by the data that we observe is identical to the population joint distribution.
- We can then just re-sample with replacement from our observed data, and see how much our estimates vary across re-samples.

Bootstrap estimation

The bootstrapping procedure is:

- Repeat many times:
 1. Take a sample of size n *with replacement* from the observed data
 2. Apply the estimating procedure on the bootstrap sample.
- Calculate the standard deviation of a parameter estimate across these many bootstrap estimates.

We can try this with the [Pager \(2003\)](#) data.

```
set.seed(60637)
B <- 200
outmat <- replicate(B,
  {
    # Take a sample of size n with replacement from the data
    idx <- sample(1:nrow(dfp), replace = TRUE)
    # fit the model on the sampled data
    lmx <- lm_robust(call_back ~ black*record,
                     data = dfp[idx,])
    coef(lmx)
  })
outmat <- t(outmat)
dim(outmat)

## [1] 200 4

head(outmat, 4)

##      (Intercept) black record black:record
## [1,]      0.37 -0.24  -0.24      0.148
## [2,]      0.37 -0.25  -0.16      0.059
## [3,]      0.36 -0.27  -0.19      0.129
## [4,]      0.34 -0.23  -0.19      0.126
```

We can try this with the [Pager \(2003\)](#) data.

```
apply(outmat, 2, sd)
```

##	(Intercept)	black	record	black:record
##	0.040	0.044	0.046	0.055

Compare this to the robust standard errors from our model.

```
model2$std.error
```

##	(Intercept)	black	record	black:record
##	0.039	0.046	0.049	0.057

We can also get confidence intervals from the bootstrap estimates.
For each parameter...

1. Sort bootstrap estimates from smallest to largest
2. Find the lower bound as the $\alpha/2$ percentile, and the upper bound as the $1 - \alpha/2$ percentile; i.e., so that $(1 - \alpha)\%$ of estimates are within this range

```
t(apply(outmat, 2, quantile, probs = c(0.025, 0.975)))
```

```
##           2.5%  97.5%  
## (Intercept)  0.267  0.414  
## black       -0.276 -0.118  
## record      -0.257 -0.093  
## black:record -0.018  0.172
```

Compare this to the confidence intervals from our model.

```
confint(model2)
```

```
##           2.5 % 97.5 %  
## (Intercept)  0.264  0.416  
## black       -0.290 -0.110  
## record      -0.270 -0.076  
## black:record -0.029  0.196
```


- `estimatr::lm_robust()` outputs robust standard errors by default; this is why it's really nice to use.
- There are options for different types of robust standard errors which have different small sample properties, but they're asymptotically equivalent.

References I

- Amrhein, V., Greenland, S., and McShane, B. (2019). Retire statistical significance. *Nature*, 567:305–307.
- Card, D. and Krueger, A. B. (1993). Minimum wages and employment: A case study of the fast food industry in New Jersey and Pennsylvania.
- Gerber, A. S. and Malhotra, N. (2008). Do statistical reporting standards affect what is published? publication bias in two leading political science journals. *Quarterly Journal of Political Science*, 3(4):313–351.
- Pager, D. (2003). The mark of a criminal record. *American Journal of Sociology*, 108(5):937–975.